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# Exercise 3: Google App Engine SDK and Web App Development

## Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

## Procedure

### Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.



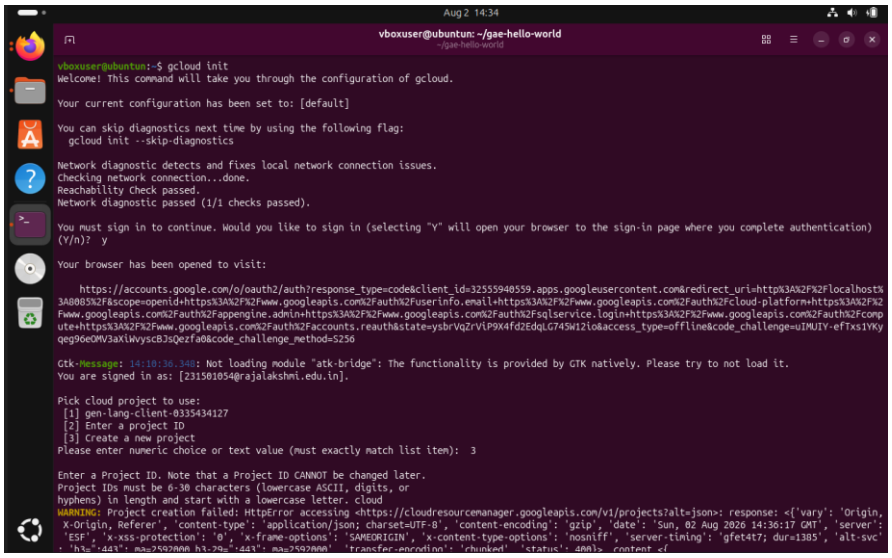

Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-linux-x86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

### Step 2: Initialize GCloud & Install Components

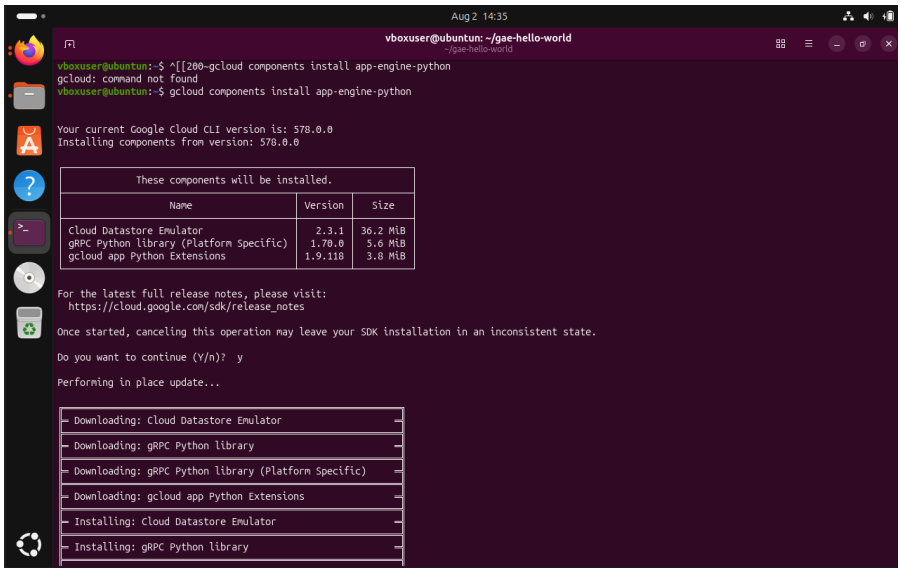
```
gcloud init
```

Sign in to your Google account when prompted.



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```
gcloud components install app-engine-python
```




## Step 3: Create the App Directory

```
mkdir gae-hello-world
cd gae-hello-world
```

## Step 4: Create Application Code (main.py)

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello World from Google App Engine!"

if __name__ == "__main__":
    app.run(host="127.0.0.1", port=8080, debug=True)
```

## Step 5: Create Configuration File (app.yaml)

```
runtime: python39
handlers:
- url: /.*
  script: auto
```

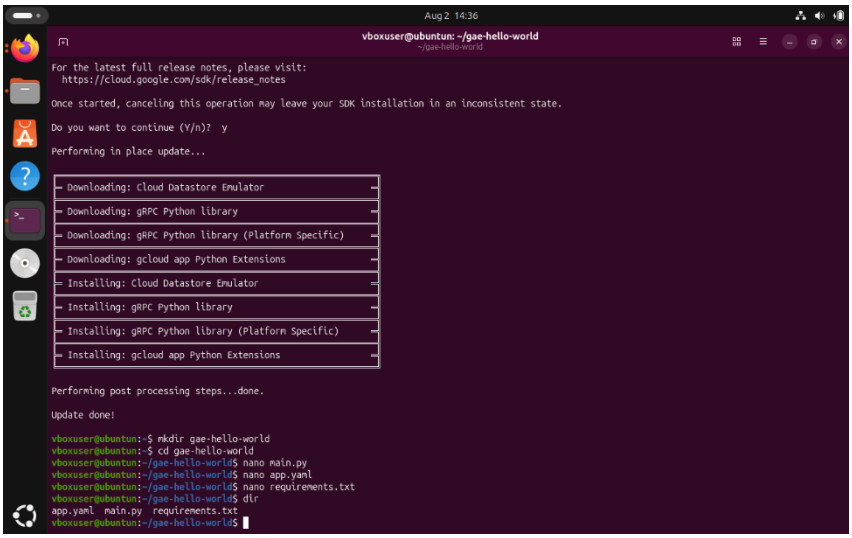
## Step 6: Create requirements.txt

```
Flask==2.0.1
```

# Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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# Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.